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Tzu-Hsien Lee

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Education

M.S. in Robotics, University of Minnesota, Twin Cities — Minneapolis, MN
Courses: Deep Learning for Robotics, Computer Vision, Robot Vision, Robotics, Optimal Estimation — 08/2026

B.S. in Mechanical Engineering, National Yang Ming Chiao Tung University — Hsinchu, Taiwan
Courses: Robotics, Object Oriented Programming, Microcomputer System and Lab, Nonlinear Control — 06/2022

Publications

- **T.-H. Lee**, F. Mahmudova, K. Desingh. "Learning Category-level Last-meter Navigation from RGB Demonstrations of a Single-instance." *IEEE Robotics and Automation Letters (RA-L)*, 2026. doi: 10.1109/LRA.2026.3707352. **[Website]**

Experience

Robotics Software Engineering Intern, AgRobotics **[Website]** — 10/2026 - Present
Skills: Robot Perception, Object Detection, VLM Auto-Labeling, Dataset Curation, TensorRT, ROS 2

- Improved strawberry detection recall by 145%, tracing the bottleneck to label noise rather than model capacity and building a VLM auto-labeling pipeline to regenerate the training set.
- Deployed the detector on NVIDIA Jetson Thor as an ONNX model integrated with the ZED stereo camera ROS 2 pipeline, meeting the real-time budget required for in-furrow operation.

Graduate Researcher, Robotics: Perception and Manipulation Lab **[Website]** — 09/2024 - 08/2026
Skills: Imitation Learning, Deep Learning, Robot Perception, Vision Foundation Models, PyTorch

- First-authored RA-L paper (2026) on category-level last-meter navigation for mobile manipulation, achieving 89.42% success on unseen instances by training goal-conditioned imitation learning policies on multi-view RGB demonstrations from a single object instance, generalizing across the entire category.
- Designed the policy architecture with a frozen DINOv2 vision encoder, language-driven SAM2 segmentation, and a score-matrix decoder that captures spatial correlation between current and goal observations for action prediction.
- Engineered an automated data collection pipeline using AprilTag-based localization to generate expert trajectories, eliminating manual teleoperation for behavior cloning.
- Developed SpotStack, a modular software library wrapping the Spot SDK for perception integration, data collection, and policy deployment on Boston Dynamics Spot.

Robotics Course Developer, CSCI 5551: Robotics, University of Minnesota **[Website]** — 01/2026 - 05/2026
Skills: Robot Manipulation, Robot Perception, 3D Vision, SAM, OpenCV, Python, ZED Stereo Camera

- Designed and engineered a 10-stage real-robot manipulation curriculum on a 6-DoF Lite6 arm and ZED 2i stereo camera, covering arm kinematics, frame transforms, and perception-to-manipulation learning.
- Built reusable software infrastructure for rapid development of vision-based manipulation pipelines, including AprilTag-based camera-to-robot calibration, grasp primitives, and camera wrapper APIs.
- Demonstrated reference solutions by implementing zero-shot target selection and 6D object pose estimation using SAM-based segmentation and oriented bounding box fitting on point clouds.

Software Lead, NYCU Autonomous Underwater Vehicle (AUV) Team **[Website]** — 02/2021 - 09/2022
Skills: C/C++, Python, ROS, Control, Sensor Fusion, Path Planning, Localization

- Architected a modular high-level/low-level robotics system by using an embedded system (STM32) for low-level attitude and motion control, and deploying ROS on Raspberry Pi and Jetson Nano for task planning and navigation.
- Implemented a gradient descent-based Madgwick filter on embedded hardware that reduced high-frequency attitude noise by 10%, and developed real-time A*-based navigation using sonar localization.

Technical Skills

Languages: Python, C, C++, MATLAB, JavaScript
Software & Tools: PyTorch, Hugging Face Transformers, ROS 1/2, TensorRT, ONNX, Gazebo, OpenCV, Git, Docker
Robotics Platforms & Systems: Boston Dynamics Spot, NVIDIA Jetson (Thor, Nano), STM32, Raspberry Pi

Projects

One-Shot Imitation Learning via Visual Servoing — 11/2024 - 12/2024

- Achieved one-shot manipulation generalization by implementing a two-stage imitation learning framework combining object-centric segmentation and learned visual servoing using PyTorch and ROS on a Kinova arm in Gazebo.
- Trained a Siamese CNN servoing policy on features extracted from a FiLM-conditioned U-Net segmentation model, enabling closed-loop end-effector control from a single visual demonstration.